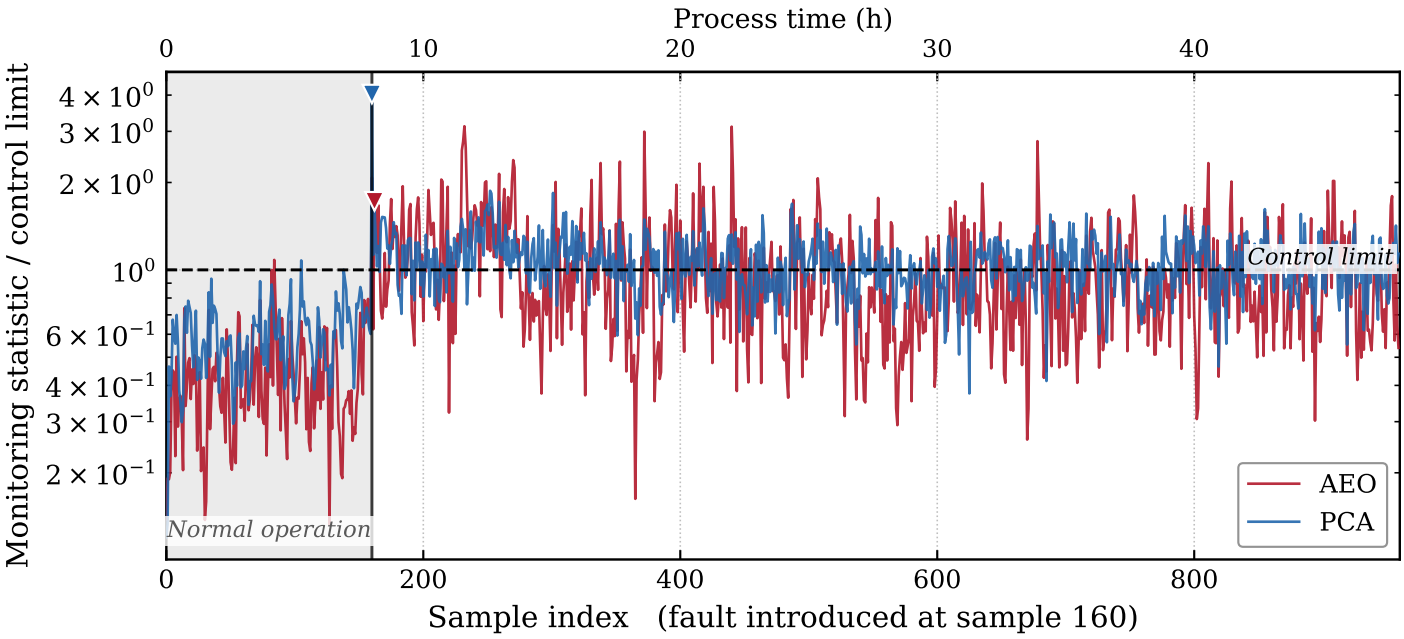
# Fault Detection Performance for IDV(4)

Hotelling's $T^2$ statistic — AEO vs. PCA at a common 1% false-alarm rate




**Held-out performance (mean of 7 independent runs)**

| | | | | |
|---|---|---|---|---|
| AEO | delay | 22.0 samples (66 min) | MDR 0.675 | detected in 7/7 runs |
| PCA | delay | 11.1 samples (33 min) | MDR 0.494 | detected in 7/7 runs |

**PCA detects 10.9 samples (33 min) earlier; PCA missed-detection rate lower by 27%**